3D HD95 per organ: boundary loss and 2.5D input

ResEnc DS U-Net · strong geometric augmentation · 50 epochs · seed 0 · 4-fold CV, 10 val patients per fold · lower is better

Esophagus

3D HD95 (mm)

15
10
5

8.5 ± 2.1
8.1 ± 2.6
9.2 ± 3.5
8.8 ± 2.6
8.4 ± 2.4

Heart

3D HD95 (mm)

20
0

11.6 ± 8.6
13.0 ± 9.3
7.9 ± 1.4
6.6 ± 1.1
8.2 ± 2.7

Trachea

3D HD95 (mm)

50
0

8.9 ± 1.8
21.8 ± 22.8
9.0 ± 2.3
8.1 ± 1.2
8.0 ± 1.3

Aorta

3D HD95 (mm)

20
10
0

10.9 ± 6.2
11.7 ± 4.0
9.5 ± 2.7
9.7 ± 4.8
6.9 ± 2.2

All organs (mean)

3D HD95 (mm)

20
0

10.0 ± 2.2
13.7 ± 6.9
8.9 ± 1.7
8.3 ± 1.7
7.9 ± 1.2

2D
no boundary (control)

2D
ramped boundary

2D
fixed boundary 0.08

2.5D
ramped boundary

2.5D
fixed boundary 0.08

One fold
Mean ± SD over the 4 folds
No boundary
Ramped boundary
Fixed boundary 0.08
2D input
2.5D input (3 slices)

3D HD95 (mm) on the original CT grid at the best (val-selected) epoch; each point is a patient-averaged fold. Large marker = mean and SD over the 4 folds. Colour = loss, shape = input. One seed and 10 patients per fold: differences of a few thousandths of Dice are within noise. Y-axes differ by organ.